

Muhammad Bilal Tahir

Artificial Intelligence Student & Software Developer

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PROFESSIONAL SUMMARY

Detail-oriented Artificial Intelligence student and software developer with documented success engineering scalable software MVPs, real-time communication architectures, and robust machine learning pipelines. Expert at designing high-reliability web ecosystems, deep learning classification targets, and automated data engines. Proactive learner specializing in computer vision, state management, and systemic cybersecurity defenses.

CORE TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL (PostgreSQL), HTML5, CSS3

AI, Vision \& Real-Time: PyTorch, YOLOv8 Engine, WebSockets (Socket.io), Next.js App Router, React

Backend \& Infrastructure: Supabase BaaS, PostgreSQL, Node.js, Express, Railway Cloud, File I/O

Development Tools: Cursor IDE, Claude Code, Git Version Control, n8n Automation Engine

FEATURED ENGINEERING PROJECTS

YOLO Object Detection Engine

Python, YOLOv8, Computer Vision, Deep Learning

AI Developer

github.com/bilaltahir1021/YOLO

- Assembled an automated deep learning configuration using custom bounding boxes and YAML metadata structures.
- Configured training frameworks executing iterative epochs to monitor loss tracking distributions and valuation metrics.
- Generated persistent analytical graphs and bounding validation models verifying item classifications natively.

Real-Time Messaging Application (ChatApp)

JavaScript, React, Node.js, Express, Socket.io

Full-Stack Developer

github.com/bilaltahir1021/Chatapp

- Engineered a full-stack real-time messaging workspace utilizing WebSocket protocol layers via Socket.io.
- Constructed an isolated backend server using Node.js and Express to manage active client connections.
- Implemented persistent message state controls on a modular React frontend ensuring instantaneous client-side UI sync.

Micro-CRM MVP (Premium Travel Platform)

Next.js, Supabase, PostgreSQL, Tailwind

Full-Stack Application Developer

github.com/bilaltahir1021/micro-crm

- Architected relational database schema tiers managed via Supabase cloud engines for strict transactional integrity.
- Constructed dynamic server action modules, App Router API paths, and a polished responsive user interface.
- Resolved state synchronization parameters across multi-tier lookup filters and persistent data instances.

Aegis Crawler (Self-Healing Scraping Framework)

n8n, Autonomous AI Logic, Workflow Orchestration

Automation Architect

Private Repository

- Designed a continuous multi-stage pipeline using the n8n logic canvas for automated mass site data collection.
- Integrated automated retry loops with dynamic failure tracking to handle layout shifts autonomously.

Sleep Consistency Analytics Engine

C++, Localized Storage Streams, Data Analytics

Systems Programmer

github.com/bilaltahir1021/Sleeptacker

- Programmed a low-level console tool utilizing native persistent File I/O disk parsing operations cleanly.
- Coded calculations outputting consistency ratings via pattern tracking metrics over chronological arrays.

EDUCATION \& ACADEMIC HISTORY

Bachelor of Science in Artificial Intelligence (BS-AI)

2023 -- 2027

Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology (PAF-IAST)

GCE Advanced Levels (A-Levels)

2020 -- 2022

Lahore Grammar School, Johar Town Campus (LGS JT)

CREDENTIALS \& CERTIFICATIONS

- **Google Cybersecurity Professional Certificate (Coursera):** Asset defense, incident response, network logs, and Linux.
- **Technical Operations:** Active lab deployment on TryHackMe alongside applications for global Erasmus+ IT internships.